Theorem 8.1 (Section Formula - Internal Division)

Let O be the origin. Let A and B be two points. Let P be the point which divides the line segment AB internally in the ratio $m : n$. If $\vec{a}$ and $\vec{b}$ are the position vectors of A and B, then the position vector $\overrightarrow{OP}$ of P is given by

$$\overrightarrow{OP} = \frac{n\vec{a} + m\vec{b}}{n+m}.$$

Proof

Since O is the origin, $\vec{a}$ and $\vec{b}$ are the position vectors of A and B, we have

$$\overrightarrow{OA} = \vec{a} \text{ and } \overrightarrow{OB} = \vec{b}.$$

Let $\overrightarrow{OP} = \vec{r}$.

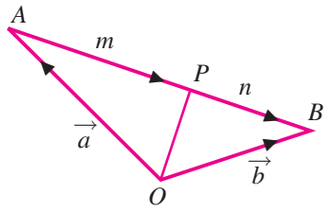


Fig. 8.26

Since P divides the line segment AB internally in the ratio $m : n$, we have,

$$\frac{|\overrightarrow{AP}|}{|\overrightarrow{PB}|} = \frac{m}{n}$$

and hence $n|\overrightarrow{AP}| = m|\overrightarrow{PB}|$.

But the vectors $\overrightarrow{AP}$ and $\overrightarrow{PB}$ have the same direction. Thus

$$n\overrightarrow{AP} = m\overrightarrow{PB}. \qquad (8.1)$$

But $\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \vec{r} - \vec{a}$ and $\overrightarrow{PB} = \overrightarrow{OB} - \overrightarrow{OP} = \vec{b} - \vec{r}$

Substituting this in (8.1), we get

$$n(\vec{r} - \vec{a}) = m(\vec{b} - \vec{r})$$

and hence

$$(n+m)\vec{r} = n\vec{a} + m\vec{b}.$$

Thus $$\overrightarrow{OP} = \frac{n\vec{a} + m\vec{b}}{n+m}.$$

Theorem 8.2 Section Formula - External Division (Without proof)

Let O be the origin. Let A and B be two points. Let P be the point which divides the line segment AB externally in the ratio $m : n$. If $\vec{a}$ and $\vec{b}$ are the position vectors of A and B, then the position vector $\overrightarrow{OP}$ of P is given by

$$\overrightarrow{OP} = \frac{n\vec{a} - m\vec{b}}{n-m}.$$

Note 8.1

By taking $m = n = 1$ in Theorem 8.1, we see that the position vector of the midpoint of the line joining the points A and B is $\frac{\vec{a} + \vec{b}}{2}$, where $\vec{a}$ and $\vec{b}$ are the position vectors of the points A and B respectively.

From the above theorem we can get a condition for three points to be collinear.

08_11th-Maths-Vol2-Unit8.indd 57 16-12-2022 11:01:08